

CAMBELL ISLAND, New ZealandWMO: **939470**Lat: **52.550S**Lon: **169.150E**Elev: **62**StdP: **14.66**Time Zone: **12.00 (NZL)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.2	32.1	22.5	16.9	33.8	24.4	18.6	35.2	42.6	40.4	39.8	40.6	14.0	230	0.982

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.0	56.9	53.5	55.1	52.2	53.9	51.2	54.5	56.1	53.1	54.5	52.0	53.2	15.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.5	61.1	55.1	52.3	58.3	53.8	51.2	56.0	52.7	23.0	56.2	22.2	54.6	21.5	53.3	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.0	35.5	32.4	26.9	61.5	2.6	3.2	25.0	63.8	23.5	65.6	22.1	67.4	20.2	69.7		
(5)			25.6	57.6	2.4	3.1	23.9	59.9	22.5	61.7	21.2	63.5	19.5	65.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	48.5	48.6	47.6	45.8	43.6	41.6	41.3	41.7	42.2	42.9	44.3	47.5
	DBStd	4.33	2.77	2.99	3.05	3.21	3.86	3.89	3.99	3.47	3.48	3.16	3.19	3.22
	HDD50	2030	61	55	86	130	198	253	271	257	233	222	173	91
	HDD65	7443	511	459	540	576	662	703	736	722	683	686	622	543
	CDD50	61	16	16	11	3	1	0	0	0	0	0	1	13
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	18.2	18.3	18.3	18.4	19.0	18.4	18.9	18.0	16.8	17.8	18.7	18.2	17.0
Precipitation	PrecAvg	52.8	4.8	4.3	4.7	4.6	5.1	4.3	4.1	4.2	4.4	4.4	3.9	4.3
	PrecMax	61.5	8.0	7.2	8.4	8.4	9.4	7.6	7.8	7.2	6.9	7.4	5.7	9.3
	PrecMin	42.2	1.7	2.3	1.5	2.1	2.3	2.2	1.5	1.6	1.2	2.3	1.5	1.2
	PrecStd	5.1	1.5	1.3	1.4	1.3	1.6	1.2	1.3	1.2	1.3	1.1	1.1	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	60.0	58.3	54.9	53.1	50.4	49.9	50.0	50.3	52.1	53.8	58.6
		MCWB	54.4	56.9	55.1	52.7	50.8	48.3	47.7	47.1	47.8	48.6	49.7	53.9
	2%	DB	55.8	56.4	55.2	53.0	51.1	48.9	48.4	48.5	48.9	50.1	51.4	55.8
		MCWB	52.2	53.7	52.9	51.4	49.5	47.3	46.7	46.3	46.9	47.3	47.8	52.0
	5%	DB	54.2	54.5	53.5	51.6	50.0	48.0	47.5	47.6	48.0	48.8	50.2	53.9
		MCWB	50.9	52.1	51.6	50.3	48.7	46.5	45.9	45.7	46.3	46.3	47.2	50.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	57.2	55.5	53.5	51.4	49.1	48.3	48.1	48.6	49.8	51.1	55.0
		MCDB	57.2	59.7	57.5	54.4	52.2	49.9	48.8	48.9	49.2	50.8	52.6	57.0
	2%	WB	53.4	54.5	53.5	51.8	50.0	47.7	47.2	46.9	47.6	48.2	49.1	53.1
		MCDB	55.1	55.9	54.8	52.5	50.7	48.5	47.9	47.9	48.3	49.4	50.3	55.1
	5%	WB	51.9	52.9	52.1	50.6	49.0	46.8	46.3	46.2	46.7	47.0	48.1	51.4
		MCDB	53.3	54.0	53.1	51.5	49.8	47.7	47.2	47.2	47.7	48.3	49.3	53.1
Mean Daily Temperature Range	5% DB	WB	50.7	51.6	50.7	49.4	48.0	46.0	45.3	45.3	45.8	45.9	47.1	50.1
		MCDB	52.3	52.7	51.8	50.4	48.9	47.0	46.5	46.5	47.0	47.4	48.6	51.8
		MCWBR	7.2	7.0	6.3	5.9	5.9	5.9	5.6	5.9	6.2	7.2	7.3	7.5
	5% WB	MCDBR	9.0	8.7	8.3	6.4	6.3	5.8	5.5	5.9	6.2	7.9	8.1	9.4
		MCWBR	6.9	7.6	7.5	6.8	6.6	6.4	5.9	5.9	6.1	7.0	6.5	6.8
		MCDBR	7.9	8.0	7.7	6.6	6.5	5.8	5.4	5.8	5.8	7.0	6.8	8.5
Clear-Sky Solar Irradiance	taub	0.341	0.339	0.335	0.328	0.316	0.310	0.312	0.325	0.352	0.343	0.337	0.340	
	taud	2.525	2.534	2.542	2.499	2.399	2.300	2.371	2.465	2.405	2.474	2.509	2.524	
	Ebn at Noon	305	295	276	243	203	178	200	239	263	290	305	309	
	Edh at Noon	33	30	26	22	17	16	17	22	29	32	33	33	
All-Sky Solar Radiation	RadAvg	1647	1300	862	488	240	152	197	404	732	1205	1554	1768	
	RadStd	106	88	68	27	19	9	17	31	50	54	65	92	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.45	N/A	+1.82	N/A	+0.95	+1.15	-162	-198	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page